

Solutions

1. Riemann tensor is of the form:

$$R_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc})$$

Apply the Second Bianchi Identity [**0.5 pt**]:

$$\nabla_e R_{abcd} + \nabla_c R_{abde} + \nabla_d R_{abec} = 0$$

Since $\nabla_e g_{ab} = 0$ [**0.5 pt**], this becomes:

$$K_{,e}(g_{ac}g_{bd} - g_{ad}g_{bc}) + K_{,c}(g_{ad}g_{be} - g_{ae}g_{bd}) + K_{,d}(g_{ae}g_{bc} - g_{ac}g_{be}) = 0$$

Contracting with $g^{ac}g^{bd}$ [**1 pt**]:

$$K_{,e}(n^2 - n) + K_{,c}(\delta_a^c \delta_e^d - n \delta_e^c) + K_{,d}(n \delta_e^d - \delta_e^c \delta_c^d) = 0$$

$$K_{,e}(n^2 - n) + K_{,e}(1 - n) + K_{,e}(n - n) = 0$$

$$(n - 1)(n - 2)K_{,e} = 0$$

$K_{,e} = 0 \implies K$ is a constant. [**simplification and final ans 1 pt**]

- 2.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$A'_\nu = \frac{\partial x^\sigma}{\partial x'^\nu} A_\sigma, \quad \partial'_\mu = \frac{\partial x^\rho}{\partial x'^\mu} \partial_\rho$$

[**1 pt**]:

$$\partial'_\mu A'_\nu = \frac{\partial x^\rho}{\partial x'^\mu} \partial_\rho \left(\frac{\partial x^\sigma}{\partial x'^\nu} A_\sigma \right)$$

$$\partial'_\mu A'_\nu = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} (\partial_\rho A_\sigma) + \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial^2 x^\sigma}{\partial x^\rho \partial x'^\nu} A_\sigma$$

Swap μ and ν [**1 pt**]:

$$F'_{\mu\nu} = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} (\partial_\rho A_\sigma - \partial_\sigma A_\rho) + \left(\frac{\partial^2 x^\sigma}{\partial x'^\mu \partial x'^\nu} - \frac{\partial^2 x^\sigma}{\partial x'^\nu \partial x'^\mu} \right) A_\sigma$$

Since the partial derivatives commute, the second term vanishes [**1 pt**].

Therefore,

$$F'_{\mu\nu} = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} F_{\rho\sigma}$$

3. The four velocity of the particle is $u^a = (1/\sqrt{1-2M/r_0}, 0, 0, 0)$, its acceleration is given by $a^b = u^a \nabla_a u^b$. This is $a_b = (0, \frac{M}{r_0^2(1-2M/r_0)}, 0, 0)$. Hence the magnitude of the force experienced by the particle due to the string is $\frac{Mm}{r_0^2(1-2M/r_0)^2}$. The energy of this particle wrt the observer at infinity is $E = -mu_a u_{\text{obs}}^a = m\sqrt{1-2M/r_0}$, where u_{obs}^a is the four velocity of the observer at infinity. Now at infinity the observer's proper length in the radial direction is identical to the coordinate r . Hence the force F_a can be calculated as $F_a dx^a = dE$. However only the radial component is nonzero. Hence we have $F_r dr = \frac{mM}{r_0^2(1-2M/r_0)^{3/2}} dr$. Hence the forces are related by the redshift factor.

0.5 for finding u^a , 1 pt for Christoffel symbols, 0.5 pt for a^b , 1 pt for explanation

4.

$$v = H_0 d = 70 \times 5.3 \times 10^3 \text{ km/s} = 3.7 \times 10^8 \text{ m/s}$$

While it appears that the relative velocity of the galaxy cluster from us is greater than c , this is not in violation of STR as this includes the effect of the expansion of the Universe, included in the space-time fabric. Further, for high distances, the Hubble's law needs modification. Recall, the derivation of Hubble's law FRWL metric.

[1 for Hubble's law, 1 for final ans, 1 for comment]

5. In the Schwarzschild metric, the energy per unit rest mass $\tilde{E}$ is a constant of motion **[1 pt]**:

$$\tilde{E} = \left(1 - \frac{2M}{r}\right) \frac{dt}{d\tau}$$

For an observer falling from rest at infinity, $\tilde{E} = 1$ **[1 pt]**. Radial geodesic equation for a massive particle simplifies to **[0.5 pt for angular piece = 0]**:

$$\left(\frac{dr}{d\tau}\right)^2 = \frac{2M}{r} \implies \frac{dr}{d\tau} = -\sqrt{\frac{2M}{r}}$$

where we choose negative root for inward motion. **[1 pt]**

$$d\tau = -\sqrt{\frac{r}{2M}} dr$$

To find the total proper time between $r_1 = 6M$ and $r_2 = 2M$:

$$\Delta\tau = \int_{6M}^{2M} -\sqrt{\frac{r}{2M}} dr = \frac{1}{\sqrt{2M}} \int_{2M}^{6M} r^{1/2} dr$$

Evaluating the definite integral:

$$\Delta\tau = \frac{1}{\sqrt{2M}} \left[\frac{r^{3/2}}{3/2} \right]_{2M}^{6M}$$

$$\Delta\tau = \frac{2}{3\sqrt{2M}} [(6M)^{3/2} - (2M)^{3/2}]$$

Final answer [0.5 pt]

$$\Delta\tau = 4M \left(\sqrt{3} - \frac{1}{3} \right)$$

CALCULATE THE FINAL NUMBER

6.

$$ds^2 = \left(1 + \frac{2\phi}{c^2} \right) dt^2 - dx^2 - dy^2 - dz^2$$

Non-zero metric components and their inverses to first order **[0.5 pt]**:

$$g_{00} = 1 + \frac{2\phi}{c^2}, \quad g_{ii} = -1$$

$$g^{00} \approx 1 - \frac{2\phi}{c^2}, \quad g^{ii} = -1$$

The Christoffel symbols are defined as:

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2} g^{\alpha\lambda} (g_{\lambda\mu,\nu} + g_{\lambda\nu,\mu} - g_{\mu\nu,\lambda})$$

To lowest non-trivial order in ϕ/c^2 :

$$\Gamma_{0i}^0 = \frac{1}{2} g^{00} (\partial_i g_{00}) = \frac{1}{2} \left(1 - \frac{2\phi}{c^2} \right) \partial_i (2\phi/c^2) \approx \frac{1}{c^2} \partial_i \phi$$

$$\Gamma_{00}^i = \frac{1}{2} g^{ij} (-\partial_j g_{00}) = \frac{1}{2} (-1) \left(-\frac{2}{c^2} \partial_i \phi \right) = \frac{1}{c^2} \partial_i \phi$$

[1 pt for Christoffel symbols]

Linearized Riemann tensor **[0.5 pt]**: $R^{\rho}_{\mu\sigma\nu} \approx \partial_{\sigma} \Gamma_{\mu\nu}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho}$.

Non-zero components **[1 pt]**:

$$R^i_{0j0} = \partial_j \Gamma_{00}^i - \partial_0 \Gamma_{0j}^i = \frac{1}{c^2} \partial_i \partial_j \phi$$

$$R^0_{i0j} = \partial_0 \Gamma_{ij}^0 - \partial_j \Gamma_{i0}^0 = -\frac{1}{c^2} \partial_i \partial_j \phi$$

Ricci tensor $R_{\mu\nu} = R^{\lambda}_{\mu\lambda\nu}$.

$$R_{00} = R^i{}_{0i0} = \frac{1}{c^2} \partial_i \partial_i \phi = \frac{\nabla^2 \phi}{c^2}$$

$$R_{ij} = R^0{}_{i0j} + R^k{}_{ikj} \approx R^0{}_{i0j} = -\frac{1}{c^2} \partial_i \partial_j \phi$$

$$R_{0i} = 0$$

Final answer: Ricci tensor components to the lowest non-trivial order in ϕ/c^2 are [**1 pt**]:

$$\begin{aligned} R_{00} &= \frac{\nabla^2 \phi}{c^2} \\ R_{ij} &= -\frac{1}{c^2} \partial_i \partial_j \phi \\ R_{0i} &= 0 \end{aligned}$$

7. First we find energy of the particle in an orbit of radius R [**1 pt**]:

$$\begin{aligned} r^2 \frac{d\phi}{d\tau} &= \ell \\ r^2 \frac{d\phi}{dt} \frac{dt}{d\tau} &= \ell \implies r^2 \Omega \frac{dt}{d\tau} = \ell \\ \frac{\ell^2}{r^4} &= \Omega^2 \left(\frac{dt}{d\tau} \right)^2 \end{aligned}$$

Using the expression for $\tilde{E}$ and multiplying by r^2 :

$$\frac{\ell^2}{r^2} = r^2 \Omega^2 \frac{\tilde{E}^2}{\left(1 - \frac{2M}{r}\right)^2}$$

From energy equation [**1 pt**]:

$$\begin{aligned} \tilde{E} &= \left(1 - \frac{2M}{r}\right) \left(1 + \frac{\ell^2}{r^2}\right) \\ \frac{\ell^2}{r^2} &= \frac{\tilde{E}^2}{\left(1 - \frac{2M}{r}\right)} - 1 = r^2 \Omega^2 \frac{\tilde{E}^2}{\left(1 - \frac{2M}{r}\right)^2} \end{aligned}$$

Rearrange to get $\tilde{E}$:

$$\tilde{E} = \frac{1 - \frac{2M}{r}}{\left(1 - \frac{2M}{r} - r^2 \Omega^2\right)^{1/2}}$$

Using Kepler's third law for the Schwarzschild geometry: $\Omega^2 r^3 = M$ [1 pt],

$$\tilde{E} = \frac{1 - \frac{2M}{r}}{\left(1 - \frac{3M}{r}\right)^{1/2}}$$

The orbit of the test particle will keep dropping with loss of energy until it reaches $6M$. As it loses more energy it plunges into the black hole as no more stable orbits exist below $6M$ [1 pt]. Now for circular orbits the potential equals the squared energy and thus

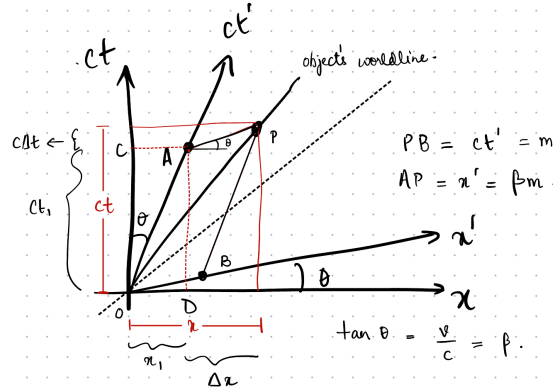
$$E(R) = \frac{R - 2M}{R^{1/2}(R - 3M)^{1/2}} \quad (1)$$

Note that the orbits occur at the stationary points of the potential, which satisfy $\frac{\partial V}{\partial R} = R^{-4}(MR^2 - L^2R + 3Ml^2)$. We can solve for the orbital angular momentum and substitute into the expression for the potential giving the above equation for the energy. Hence the difference in energy from r_0 to $6M$ will be the energy of gravitational waves emitted

$$E_{GW} = E(r_0) - E(6M) \quad (2)$$

$$= \frac{r_0 - 2M}{r_0^{1/2}(r_0 - 3M)^{1/2}} - (8/9)^{1/2} \quad (3)$$

[1 pt for final ans] No gravitational waves will be emitted in the plunge phase.



8. (a) [1 pt for spacetime diagram.]

Goal: To find $u = \frac{x}{t}$.

Coordinates of object's position P in S' , (ct', x) , can be obtained using the parallelogram OAPB.

$$u' = \frac{x'}{t'} \implies x' = u't' = \beta_1 ct'$$

$$\text{where } \beta_1 \equiv \frac{u'}{c}$$

Let $ct' = m$ for convenience. $AP = x' = \beta m$

$$\begin{aligned}(ct_1, x_1) &= (m \cos \theta, m \sin \theta) & [\mathbf{1pt}] \\ (c\Delta t, \Delta x) &= (\beta_1 m \sin \theta, \beta_1 m \cos \theta) & [\mathbf{1pt}] \\ (ct, x) &= (ct_1 + c\Delta t, x_1 + \Delta x) & [\mathbf{0.5pt}] \\ &= (m \cos \theta + \beta_1 m \sin \theta, m \sin \theta + \beta_1 m \cos \theta)\end{aligned}$$

[**0.5 pt** for substituting and final expression]

$$\begin{aligned}\frac{u}{c} &= \frac{x}{ct} \\ &= \frac{\sin \theta + \beta_1 \cos \theta}{\cos \theta + \beta_1 \sin \theta} \\ &= \frac{\tan \theta + \beta_1}{1 + \beta_1 \tan \theta} \\ &= \frac{\beta + \beta_1}{1 + \beta \beta_1}\end{aligned}$$

Finally, we obtain the formula for addition of velocities as:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

- (b) Speed of light w.r.t medium, $u' = c/n$. Using addition of velocities derived in part (a), speed of light measured by observer in lab frame:

$$u = \frac{\frac{c}{n} + v}{1 + \frac{v}{cn}}$$

[**0.5 pt** for obtaining above expression.]

For low speeds $v \ll c$, first order approximation (using binomial expansion):

$$u \simeq \left(\frac{c}{n} + v\right) \left(1 - \frac{v}{cn}\right) \simeq \frac{c}{n} + v \left(1 - \frac{1}{n^2}\right)$$

[**1 pt** for obtaining u to first order in v . **0.5** for comment on why this isn't a Galilean transformation]

9. (a) [**1 pt** for u^α, p^α] $u^\alpha = \gamma(c, \mathbf{v})$; $u_\alpha = \gamma(c, -\mathbf{v})$
 $p^\alpha = mu^\alpha = (E/c, p_x, p_y, p_z)$

Spatial Components ($\alpha = 1, 2, 3$) $\alpha = 1$ (x-component) of the equation:

$$\frac{dp^1}{d\tau} = q(F^{10}u_0 + F^{11}u_1 + F^{12}u_2 + F^{13}u_3)$$

Substituting the components [0.5 pt] :

$$\gamma \frac{dp_x}{dt} = q \left[\left(\frac{E_x}{c} \right) (\gamma c) + 0 + (-B_z)(-\gamma v_y) + (B_y)(-\gamma v_z) \right]$$

$$\gamma \frac{dp_x}{dt} = \gamma q [E_x + (v_y B_z - v_z B_y)]$$

Recognizing the cross product $(\mathbf{v} \times \mathbf{B})_x$ [1 pt]:

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

This is the classical Lorentz 3-force law [0.5 pt for final ans.].

The $\alpha = 0$ component:

$$\frac{dp^0}{d\tau} = q(F^{01}u_1 + F^{02}u_2 + F^{03}u_3)$$

$$\gamma \frac{d(E/c)}{dt} = q \left[\left(-\frac{E_x}{c} \right) (-\gamma v_x) + \left(-\frac{E_y}{c} \right) (-\gamma v_y) + \left(-\frac{E_z}{c} \right) (-\gamma v_z) \right]$$

Multiplying through by c/γ :

$$\frac{dE}{dt} = q(E_x v_x + E_y v_y + E_z v_z) = q(\mathbf{E} \cdot \mathbf{v})$$

recovers the work-energy theorem

(b) ($r > R$) [1 pt]: $Q_{encl} = \rho(\pi R^2 L)$:

$$E(2\pi r L) = \frac{\rho \pi R^2 L}{\epsilon_0} \implies \mathbf{E} = \frac{\rho R^2}{2\epsilon_0 r} \hat{r}$$

$$E_x = \frac{\rho R^2}{2\epsilon_0} \frac{x}{\sqrt{x^2 + y^2}} \quad E_y = \frac{\rho R^2}{2\epsilon_0} \frac{y}{\sqrt{x^2 + y^2}}$$

In the rest frame, there is no current, the magnetic field is:

$$\mathbf{B} = 0$$

Therefore, in the rod's rest frame,

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & 0 \\ E_x/c & 0 & 0 & 0 \\ E_y/c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (4)$$

Let the velocity of the rod w.r.t the moving frame be v along the z -axis. The boost Λ^μ_ν is:

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & 0 & 0 & -\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta\gamma & 0 & 0 & \gamma \end{pmatrix} \quad (5)$$

where $\beta = v/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$.

$$F'^{\mu\nu} = \Lambda^\mu_\rho \Lambda^\nu_\sigma F^{\rho\sigma} \text{ [1 pt].}$$

$$\mathbf{E}' = \gamma(E_x \hat{i} + E_y \hat{j}) \quad (6)$$

$$\mathbf{B}' = \frac{\gamma v}{c^2}(-E_y \hat{i} + E_x \hat{j}) \quad (7)$$

[Final ans: 1 pt]

10. (a) finding the distance between a point P (x_0, y_0) on the plane to a point on the x-axis ($x_0, 0$).

Since $dx = 0$,

$$ds^2 = \frac{dy^2}{y^2}$$

Integrating from ϵ to y_0 ,

$$d = \int_\epsilon^{y_0} \frac{dy}{y} = \ln \frac{y_0}{\epsilon}$$

As $\epsilon \rightarrow 0$, $d \rightarrow \infty$

- (b) Non-zero Christoffel symbols [2 pt = 1.5 pt for each non-zero and 0.5 pt for stating that the rest are zero]:

$$\begin{aligned} \Gamma_{xy}^x &= \Gamma_{yx}^x = -\frac{1}{y} \\ \Gamma_{xx}^y &= \frac{1}{y} \\ \Gamma_{yy}^y &= -\frac{1}{y} \end{aligned}$$

Substituting these into the geodesic equation [1 pt (0.5 pt for each correct eqn)]

$$\ddot{x}^\alpha + \Gamma_{\mu\nu}^\alpha \dot{x}^\mu \dot{x}^\nu = 0:$$

$$\ddot{x} - \frac{2}{y} \dot{x} \dot{y} = 0 \quad (8)$$

$$\ddot{y} + \frac{\dot{x}^2}{y} - \frac{\dot{y}^2}{y} = 0 \quad (9)$$

(c)

$$\frac{\ddot{x}}{\dot{x}} = \frac{2\dot{y}}{y} \implies \ln \dot{x} = \ln y^2 + \ln C$$

$$\dot{x} = Cy^2$$

$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 1$ gives us:

$$\frac{\dot{x}^2 + \dot{y}^2}{y^2} = 1 \implies \dot{x}^2 + \dot{y}^2 = y^2$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{\dot{y}}{\dot{x}}\right)^2 = \frac{y^2 - \dot{x}^2}{\dot{x}^2}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{y^2 - C^2y^4}{C^2y^4} = \frac{1 - C^2y^2}{C^2y^2}$$

$$dx = \frac{Cy}{\sqrt{1 - C^2y^2}}dy$$

$$x - x_0 = \int \frac{Cy}{\sqrt{1 - C^2y^2}}dy$$

$$u = 1 - C^2y^2, \quad du = -2C^2ydy$$

$$x - x_0 = -\frac{1}{C}\sqrt{1 - C^2y^2}$$

$$(x - x_0)^2 = \frac{1}{C^2}(1 - C^2y^2) = \frac{1}{C^2} - y^2$$

$$(x - x_0)^2 + y^2 = R^2$$

where $R = 1/C$.